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Vellore Institute of Technology
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**B.Tech/ M.Tech
Capstone Project**

A Trust-Aware, Energy-Efficient Acoustic Event Reasoning Pipeline for Underwater Acoustic Sensor Networks

Team members:

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Faculty guide :

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Aim

- To design, build, and validate a trust-aware, energy-efficient acoustic event reasoning pipeline for underwater acoustic sensor networks (UASNs) — combining event-level confidence scoring, causal explanation, and confidence-gated escalation — entirely in simulation on public acoustic datasets.

Abstract

- Underwater acoustic sensor networks (UASNs) typically decide whether to "wake" on an acoustic event using a fixed spectral-energy threshold, and existing trust models in this space evaluate whether a sensor node is compromised or faulty rather than whether an individual acoustic event is genuine. This project addresses that event-level gap directly.
- We propose a three-module pipeline: (1) an Event-Level Acoustic Wake Trust Engine that produces a calibrated per-event confidence score from spectral, temporal, and simulated sensor-health signals; (2) a compact Marine Acoustic Causal Reasoning knowledge graph that generates a short explanation chain for each classified event; and (3) a Confidence-Gated Energy-Aware Escalation policy that invokes the causal-reasoning module only for high-trust events, balancing detection accuracy against simulated energy cost. All modules are built and evaluated in simulation on public datasets such as DeepShip, at TRL-4.

Literature Review

Area	Existing work	Gap identified
UASN Digital Twins (Song et al., 2024)	Digital Twin-based Network Architecture (DTNA) focusing on MARL data-patching and resource allocation.	Lacks predictive void-hole avoidance; existing void-avoidance protocols remain strictly reactive.
Digital Twin Asset Monitoring (Chen et al., 2024)	Digital twin dynamic modeling effectively applied to transmission line ice-shedding/deicing systems.	Has not been adapted for marine environments (e.g., Ice-Keel-Aware Digital Twin Link Prediction for UASNs).
Passive Acoustic Monitoring (Kammegne et al., 2023)	Passive acoustic monitoring predominantly uses deep learning solely for classification tasks.	Operates purely on pattern recognition; lacks event-level causal explanation or trust scoring.
Marine Acoustic Reasoning (Domain Review, 2025)	Identifies causal reasoning and physical knowledge integration as key emerging needs in underwater acoustics.	Marine Acoustic Knowledge Graphs (KG) and Counterfactual Verification mechanisms are largely absent and open research gaps.

Research Gap

- **Event-Level Trust & Escalation is Unaddressed:** Existing UASN trust literature scores nodes (is this sensor compromised or faulty?), not individual acoustic events (is this specific blip a genuine wake trigger?). Furthermore, while multi-objective WSN scheduling is prior art, no narrow, defensible policy gates expensive causal reasoning directly on event-level trust.
- **Absence of Causal Knowledge-Graph Reasoning in UASNs:** Causal-KG methods and audio-specific KGs (e.g., iKnow-audio) exist in other domains, but neither has been adapted to marine acoustic event explanation. Existing deep learning classifiers lack counterfactual verification.
- **Lack of Hardware-Level Gating & Adaptive Normalization:** While sensor fusion exists, explicitly separating and executing flow-acoustic cross-modal turbulence gating (using flow state to swap software pipelines) remains unexplored. Similarly, SSP inversion is mature for macro-scale navigation but undocumented as a live edge-classifier normalization signal.
- **Siloed Digital Twin Applications:** Digital twins for physical asset hazards (like ice) exist in power infrastructure, and UASN twins exist for open water, but nobody has combined the two to solve ice-keel-aware link prediction.

Objectives

- **Objective 1:** Design and train a calibrated, event-level trust-scoring model (Bayesian / evidential) that flags low-confidence acoustic wake events, benchmarked against a fixed-threshold baseline using ROC, false-alarm-rate, and calibration analysis. Contribution: Introduces event-level gating vs. traditional node-level trust.
- **Objective 2:** Build a compact causal knowledge graph linking acoustic event type to plausible causes (species, vessel class, environmental activity) and generate human-readable reasoning chains for classified events. Contribution: Addresses the domain gap by providing Marine Acoustic KG causal verification.
- **Objective 3:** Implement a confidence-gated escalation policy that invokes causal reasoning only for high-trust events, and quantify the accuracy-vs-simulated-energy trade-off against an always-on baseline.
- **Objective 4:** Validate the complete pipeline end-to-end in simulation on public acoustic datasets (e.g., DeepShip), consistent with a TRL-4 scope.

References
[1] "A Digital Twin-based Intelligent Network Architecture for Underwater Acoustic Sensor Networks," arXiv preprint arXiv:2410.20151, 2024.

[2] Chen et al., "Research on Digital Twin Dynamic Modeling Method for Transmission Line Deicing System," Energies, vol. 17, no. 6, p. 1424, 2024.

[3] Kammegne et al., "Passive acoustic monitoring uses deep learning for classification," 2023.

[4] Domain Review, "Causal reasoning and physical knowledge integration in underwater acoustics," 2025.

Sustainable Development Goals: Goal 9 (Industry, Innovation and Infrastructure), Goal 14 (Life Below Water).

Expected Outcomes: Patent / Conference (Scopus) / Journal (Scopus) / Start-up/ Product/ Others. Technology Readiness Level: Experimental Proof of Concept (TRL-3) progressing to Technology Validated in Lab

(TRL-4).